

Grammar of Graphics

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Data visualizations are made up of layers. Each layer consists of three parts:

data	tabular dataset associated with the layer
geom	graphical element associated with each observation
aesthetics (aes)	mappings from properties of the plot that associate features in the dataset with elements of the geometry

Complex plots can be constructed by putting together multiple layers.

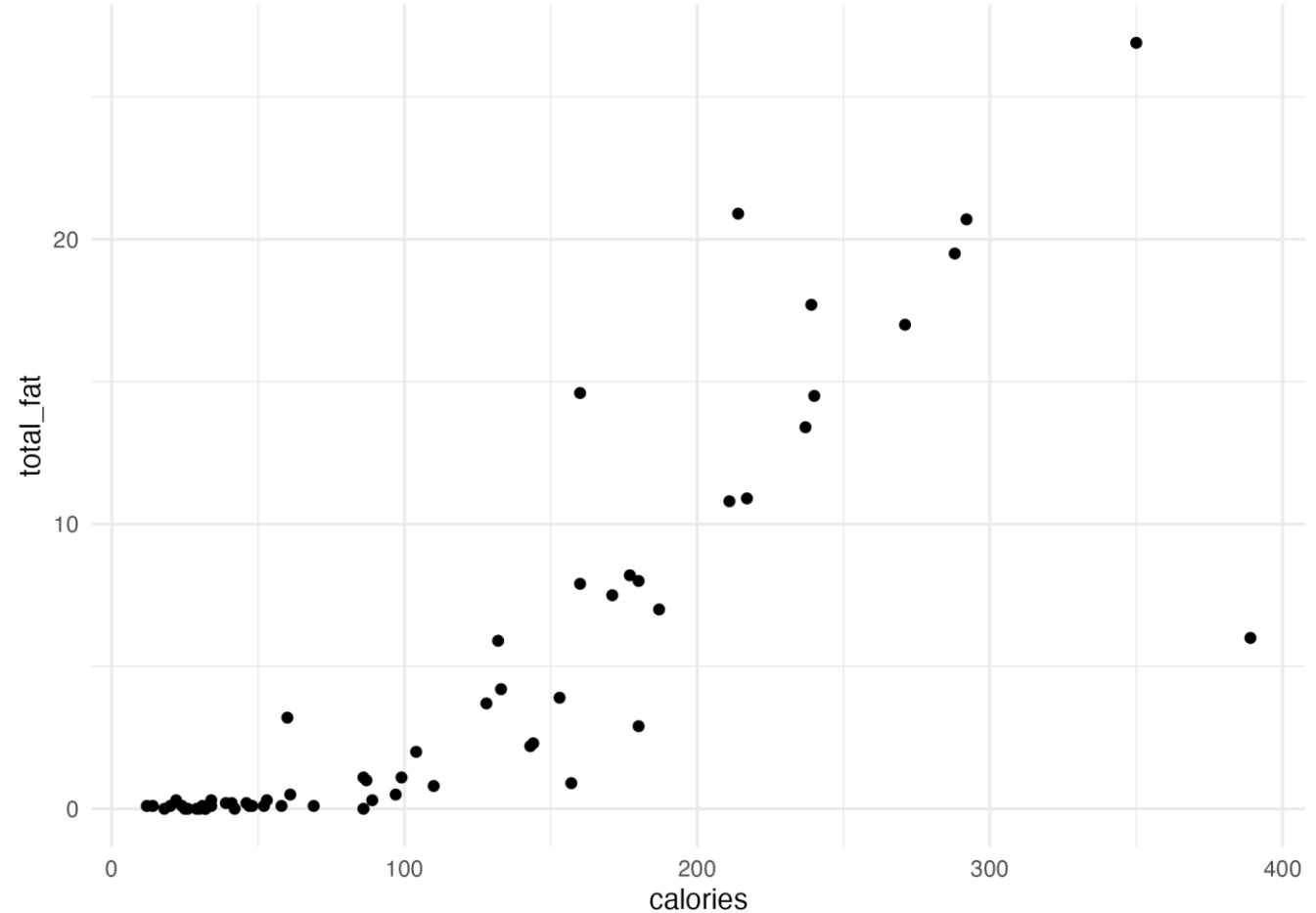
Scatter Plot Example

data: food

geom: point

aes: { x: calories, y: total_fat }

each **observation** in the
dataset is represented by a
point



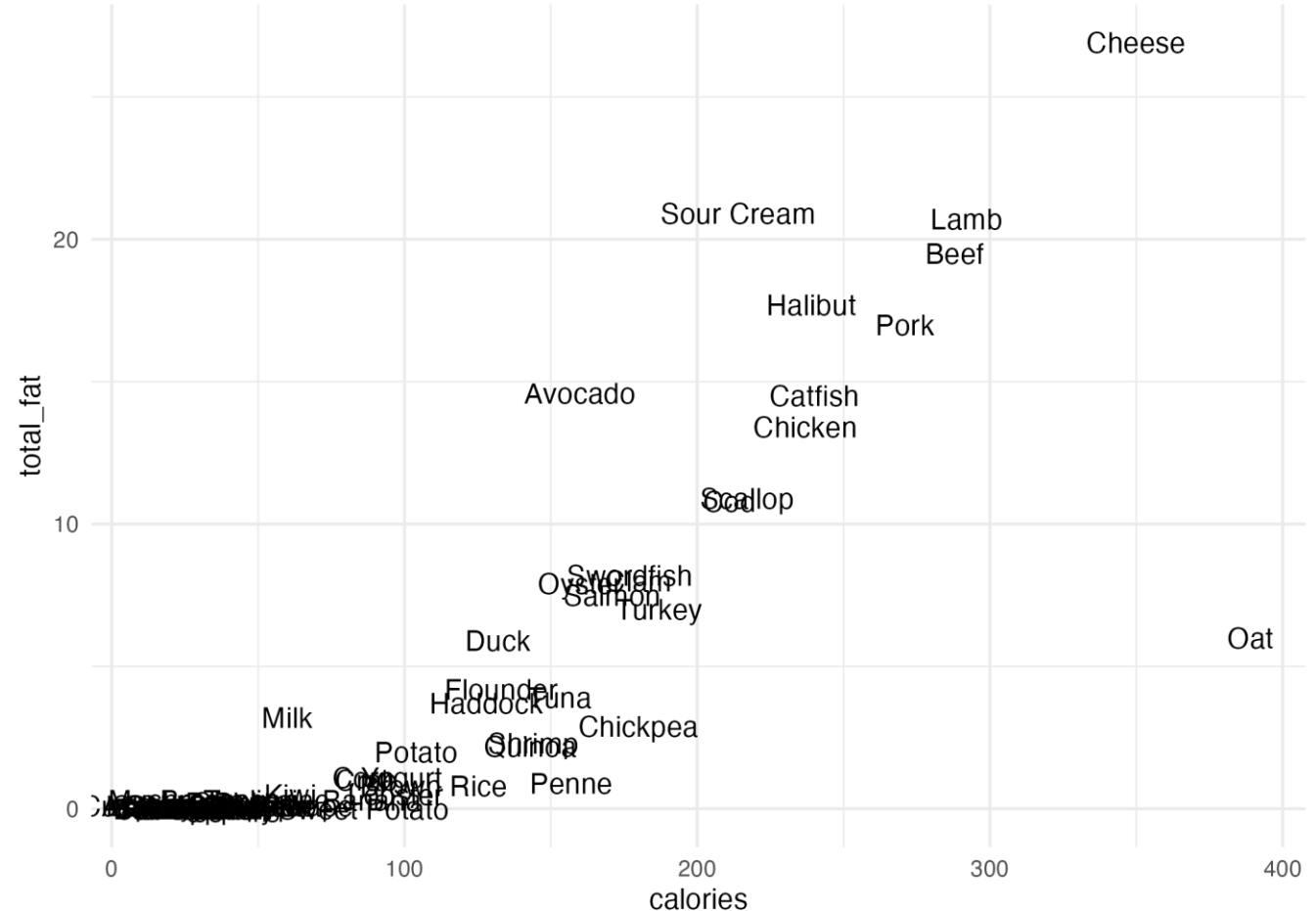
Text Plot Example

data: food

geom: text

aes: { x: calories,
y: saturated_fat
label: item }

each **observation** in the
dataset is represented by a
textual label



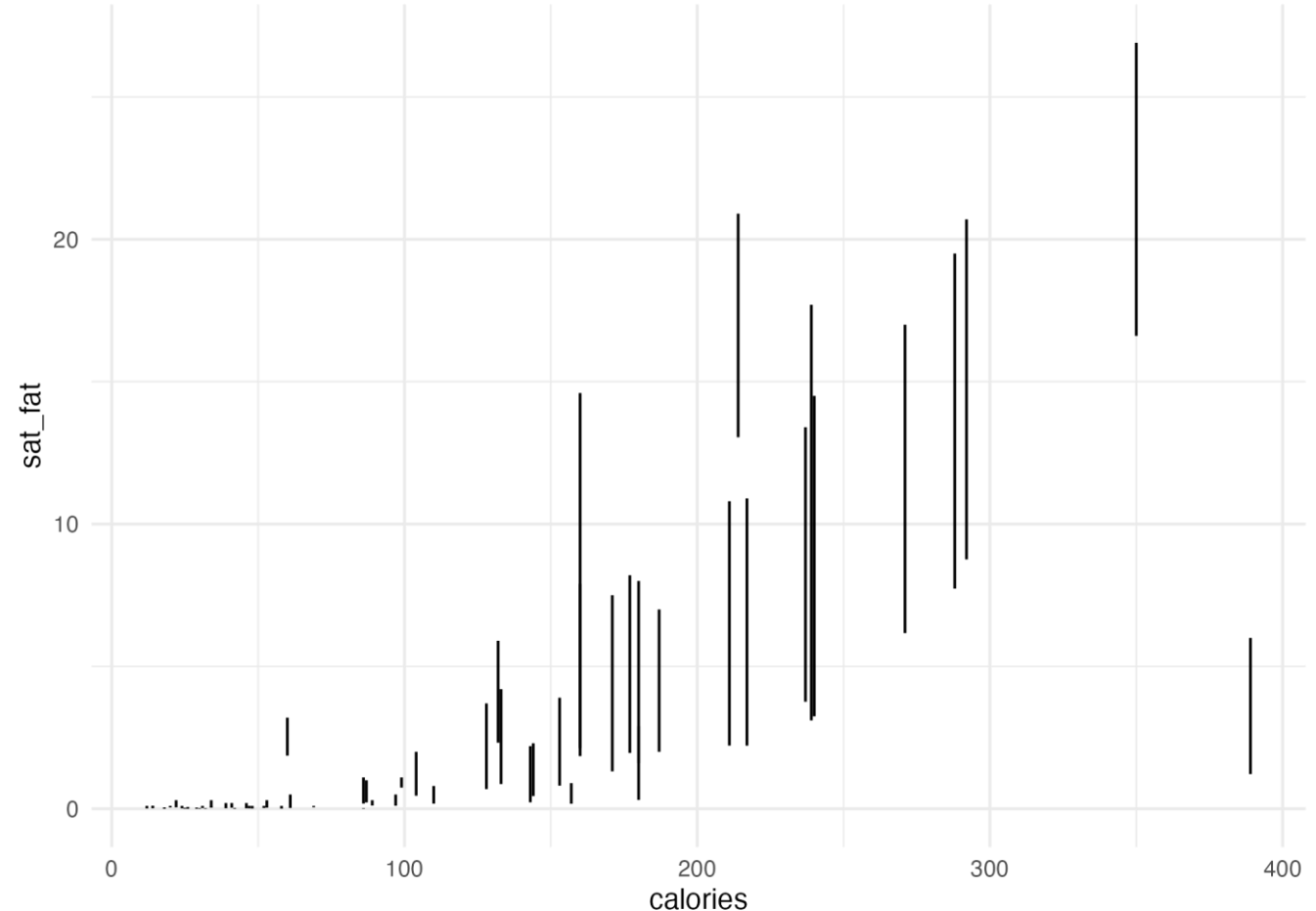
Segment Plot Example

data: food

geom: segment

aes: { x: calories,
y: saturated_fat
xend: calories,
yend: total_fat }

each **observation** in the dataset is represented by a vertical line segment



Arrow Plot Example

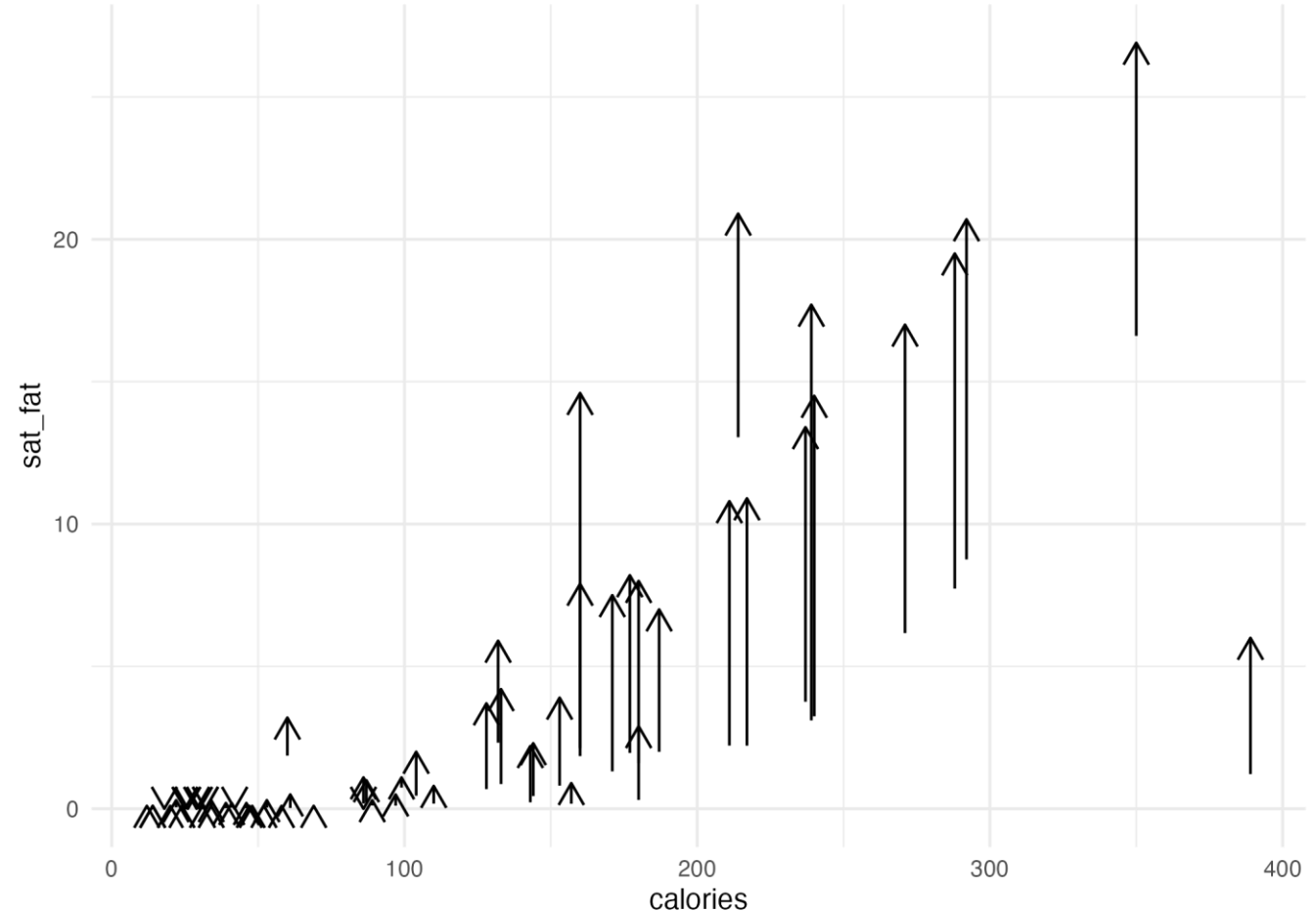
data: food

geom: arrow

aes: { x: calories,
y: saturated_fat
xend: calories,
yend: total_fat }

each **observation** is
represented by a vertical line
segment plus an arrow!

[we'll see how to do these shortly]



Syntax

```
data: food  
geom: point  
aes: { x: calories, y: total_fat }  
  
(ggplot(food) +  
  geom_point(aes(x = "calories", y = "total_fat")))
```

```
data: food  
geom: text  
aes: { x: calories,  
        y: saturated_fat  
        label: item }  
  
(ggplot(food) +  
  geom_text(aes(x = "calories", y = "total_fat", label = "item")))
```